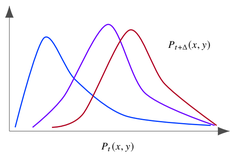
Deployed predictive system in a changing environment

Transportation · Health · Finance · Industry · IoT · Energy · Activity · Occupancy · ...

Why maintenance is challenging in practice?

Data and environment change over time



Feedback is scarce and costly

Only a small fraction of labels can be queried

Multiple interventions are available

- HOLD
- Sparse repair
- Evidence escalation
- Full retraining
- Full + repair

Intervention may introduce risk

A repair can harm probability quality and operational behaviour

Operational behaviour must be preserved

Decision · Ranking · Confidence · Calibration

These outputs must remain unchanged or controlled

Goal: autonomous maintenance

Decide when, how, and how strongly to intervene, using limited feedback while controlling risk and preserving the deployed system.

VARIEL

Three-cabinet decision architecture for predictive-system maintenance

Inputs from deployment

Current deployment window W_t

- Unlabeled target instances
- Current operating context

Actionable history $\mathcal{H}_{t-1}$

- Past queried labels
- Past maintenance outcomes
- Temporally filtered evidence

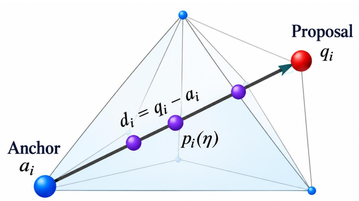
Admissible operating regimes ρ

- Frozen source state
- Periodic state
- Rolling full-retraining state
- Repaired state

Temporal Actionability

- Stale labels may identify opportunity
- Only actionable evidence may justify action

Probability-Action Geometry (V-PAG)



$$p_i(\eta) = (1-\eta)a_i + \eta q_i$$
$$0 \le \eta \le 1$$

- $\eta = 0$: HOLD
- $0 < \eta < 1$: partial repair
- $\eta = 1$: full proposal

Gain geometry

$G(\eta) = 2D\eta - C\eta^2$

Prospective liability

$H_t(\eta) = A_t\eta^2 + 2B_t\eta$, $H = 0.02$

V-RAR

(VARIEL Residual Action Refinement)

- Takes any admissible state as anchor
- Constructs a frozen residual proposal
- Acts under audit, liability and preservation constraints

Resource Cabinet

How much evidence should be bought before committing?

- **ASPIRE-CB** — Immediate certificationof easy states
- **PPARC** — Prediction-poweredaction-resolution
- **SHARP** — Variance-awaresequential allocation
- **CPPI-GREG** — Cross-prediction-regression refinement
- **RACS** — Residual actionconsensus synthesis

Easy states resolve early

Hard states escalate deeper

1080 / 1080 operational decisions

≈ 2.35% mean labels

Exact winner identity and operational equivalence are separated by design.

Maintenance Action Cabinet

What should the system do now?

- **HOLD** — No intervention is a validdecision
- **Sparse REPAIR** — Probability update withlimited current labels
- **Evidence ESCALATION** — Acquire more feedbackbefore committing
- **FULL** — Retrain on all available data
- **FULL + REPAIR** — Retrain and then repairprobabilities

Geometry-stratified canary audit

$B = \lfloor 0.04 N_{\text{target}} \rfloor$

$n_h \propto N_h R_h$

- Disjoint audit labels
- Estimate $\widehat{D}_t, D_{L,t}, D_{U,t}$
- Compute η_{RO}

Safeguard Cabinet

How far may the system act, and what must remain protected?

1 Exact prospective liability

$$H_t(\eta) = A_t\eta^2 + 2B_t\eta$$
$$\eta_H(0.02)$$

- Exact safe-action frontier
- Clip requested action to satisfy $H \le 0.02$

2 High-information guard

1% ⟶ 2% ⟶ 4%

Dominance Guard

$\eta_{\text{dom}} = \min\{\eta_{\text{RO}}, 2D_L/C\}$

Quadratic Ambiguity Half-Step

$\eta_{\text{amb}} = \frac{1}{2}\eta_{\text{RO}}$

- Dominant → guarded repair
- Unresolved positive near-tie → half-step

80 / 80 strict wins at equal eventual 100% labels

3 Preservation Guard

- Hard decisions
- Weak ranking order
- Confidence order
- Calibration-bin means

ECE-invariant projection

$r_\perp = [I - A^\top (AA^\top)^+ A]r$

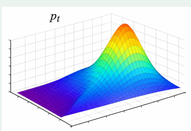
Preserves declared invariants
Retains nonzero residual correction

24 / 24 final probability-quality cells

zero protected-metric change

Guarded deployment output

Repaired probability state p_t



Selected maintenance state

- HOLD
- Sparse REPAIR
- Evidence ESCALATION
- FULL
- FULL + REPAIR

Protected operating outputs

Decision · Ranking · Confidence · Calibration

Causal label reuse

- Current canaries audit the present action
- Labels enter history only after action is fixed

Causal update to $\mathcal{H}_t$ after audited action